

ECM Flash Procedure (wiTECH 2)

Check the engine computer calibration and update it if a newer one exists

2018 Alfa Romeo Stelvio 2.0T (GU), 2.0L GME-T4, sales code EC2 | VIN ZASFAKPN5J7B88115

Where this comes from

The steps are transcribed from FCA's own dealer instructions for this exact car: **Customer Satisfaction**

Notification W05 (March 2020, 2018 GU Stelvio 2.0L EC2: "Reprogram PCM, perform PROXI Alignment and Phonic Wheel Replacement routine") and **TSB 18-026-20** (March 25, 2020, 2018 GU Stelvio 2.0L EC2, North America). Items marked *[practice]* are general flashing precautions, not from those documents.

TSB 18-026-20's fix list includes two EVAP-related codes: **P1CEA** (boost-side EVAP purge performance) and **P24D6** (EVAP system pressure sensor/switch circuit). No FCA bulletin publishes calibration numbers; wiTECH shows the current and newest part numbers on the Flash tab (step 7), which is the only way to know if this car is current.

Installed now (read from this car)

Field	Value
ECU software number	52170619
Supplier software	P235QB39, version 0000
Hardware	MM10JAHW232 (FIAT drawing 52055320)
ECU spare part number	50544870

Before you start

- **Battery support is mandatory.** Charger set so the battery stays between **13.2 and 13.5 V** for the whole flash; set any charger timer to continuous. Measure with an accurate stand-alone voltmeter, not the charger's meter. If the voltage is too high, switch on the park lamps, headlamps or HVAC blower to add load. (W05 step 1; TSB 18-026-20)
- **Do not interrupt the flash.** If it is aborted or interrupted, restart it. (W05; TSB 18-026-20)
- *[practice]* Laptop on mains power with sleep and hibernate disabled; stable, wired internet (wiTECH 2 is web-based); close other diagnostic software.
- *[practice]* Unplug the vLinker and close MultiEcuScan and cuore's live link: only the wiTECH interface may be on the OBD port.
- *[practice]* Doors closed, lights and accessories off unless used for load, key fob in the car, park brake on.

Interface: W05 lists the wiTECH MicroPod II, a laptop and wiTECH software as the special tools. If you use another interface, confirm wiTECH lists it as supported for flashing before you start.

Procedure

1. Battery support

Open the hood. Connect the charger and confirm 13.2 to 13.5 V at the battery with a stand-alone voltmeter.

2. Connect

Connect the wiTECH interface to the OBD data link connector.

3. Ignition RUN

Place the ignition in RUN (engine off).

4. Sign in

Open wiTECH 2 and sign in with your account. (W05 shows the dealer sign-in: user ID, password and dealer code, then Accept.)

5. Select the vehicle

On the Vehicle Selection screen select this car. Confirm the VIN reads ZASFAKPN5J7B88115.

6. Open the ECM

On the Topology screen click the **ECM** icon. (2018 Stelvio: PCM only; W05 adds the TCM only for 2019 cars.)

7. Compare part numbers: this answers the version question

On the ECM screen select the **Flash** tab and compare the **Current ECU Part Number** with the **New ECU Part Number**. Write both down on the record sheet.

- **Same:** the ECM is already on the latest calibration. Do not flash. Go to step 16 (W05: "proceed to Step 22"). The software lead is closed.
- **Different:** a newer calibration exists. Continue with step 8.

8. Accept and flash

On the flash agreement page tick the box, select **Flash ECU**, and follow the on-screen instructions to completion. Expected flash time is about 8 minutes (TSB 18-026-20). Watch the voltmeter throughout.

9. PROXI Alignment (BCM)

From Topology select **BCM**, then **Misc. Functions**, and run **PROXI Alignment**.

10. View and clear DTCs

When the flash is complete select **View DTCs**. From Topology go to **Action Items**, **All DTCs**, **Clear All DTCs**. The flash sets codes in other modules; that is expected.

11. Phonic Wheel Replacement (ECM)

From Topology select **ECM**, **Misc. Functions**, **Phonic Wheel Replacement**, and follow the prompts to completion.

If this routine is not done correctly, **DTC P1300 Flywheel Self Learning** will stay active. (W05; TSB 18-026-20)

12. Clear DTCs again

Topology, **All DTCs**, **Clear All DTCs**.

13. Confirm the new part number

Re-open the ECM **Flash** tab and confirm the Current ECU Part Number now equals the New ECU Part Number. Record it.

14. BCM software

TSB 18-026-20: the BCM must be at the latest available software after this repair. On the BCM Flash tab compare current and new part numbers the same way, and update if different.

15. Record identity

[practice] After sign-out, a MultiEcuScan read of the engine ECU records the new Software number in its log, so cuore picks up the new version automatically.

16. Finish

Ignition OFF, disconnect the wiTECH interface, remove the charger, close the hood.

After the flash: checking the EVAP result

- Drive normally over several days: fuel between about 15 and 85 percent, and include cold starts after the car has sat overnight. The leak tests only run under those conditions.
- Then read the ECM: cuore's **verify_repair** for P0455, P0456 and P0440, or an MES scan. All three showing **passed since clear** means the calibration fixed it.
- The check-engine light cannot confirm it on this car: the ECM stored these codes without requesting the light.
- If the part numbers matched in step 7, or the codes return after the flash, the next suspects are the ESIM part number fitted (04861961AD, never checked against the VIN) and the ESIM-to-ECM wiring.

Record sheet

Item	Result
Date / odometer	
Battery voltage during flash (min / max)	
ECM Current ECU Part Number (before)	
ECM New ECU Part Number	
Flash performed? (yes / not needed)	
PROXI Alignment completed	
Phonic Wheel Replacement completed	
DTCs present after flash (before clearing)	
ECM part number after flash	
BCM current / new part number	
Notes	

Sources: FCA US LLC Customer Satisfaction Notification W05, Dealer Service Instructions, March 2020 (static.nhtsa.gov/odi/tsbs/2020/MC-10175111-9999.pdf); TSB 18-026-20, March 25, 2020 (static.nhtsa.gov/odi/tsbs/2020/MC-10174245-9999.pdf).